KnackELT Architecture

How knack-elt works, and where it sits in a full Knack → warehouse → BI stack.

Four diagrams:

> **Scope note.** This repo is the **ingestion** piece only. The dbt models, BI configuration, and CI orchestration shown in diagram 1 are drawn from a working production deployment where they live in a companion application repo (whose `pipelines/knack_dlt.py` is the production twin of this repo's pipeline). They are real, not aspirational — but they are not in this tree. Subgraph labels mark what lives where.
>
> An adjacent, separate consumer path (LanceDB / RAG / chat) is described in RAG_ARCHITECTURE.md and is deliberately not folded in here.

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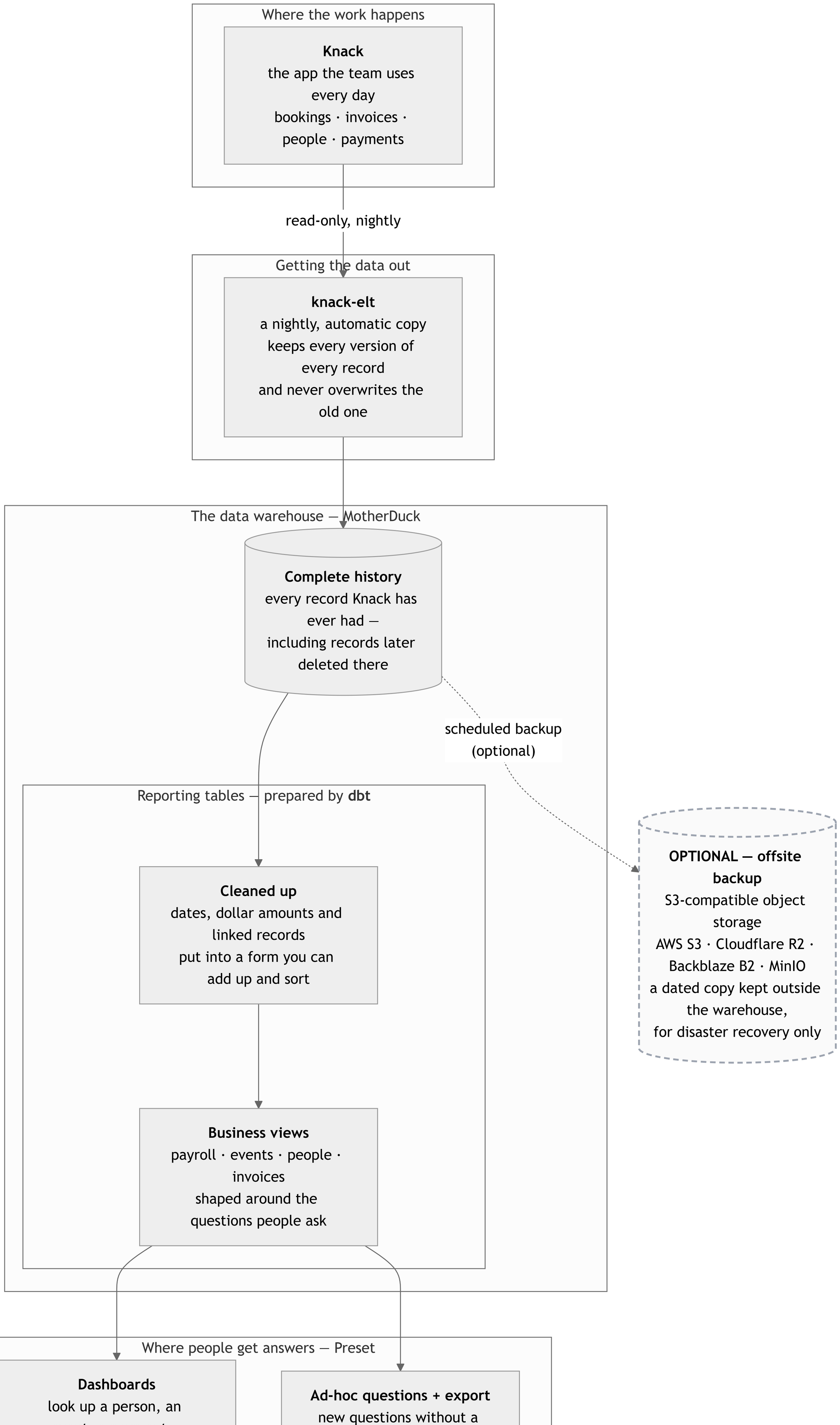
1. Reference architecture

Knack is the system of record and the operational UI. It is not a warehouse: no SQL, no aggregates, no history, and a record cap that eventually forces pruning. The stack below turns it into one — MotherDuck as the warehouse, dbt as the transformation layer, Preset (managed Apache Superset) as the access layer.

Same architecture, drawn twice: once for a stakeholder audience, once with the detail an engineer needs.

The stack in plain language

FIGURE 1 — THE STACK IN PLAIN LANGUAGE

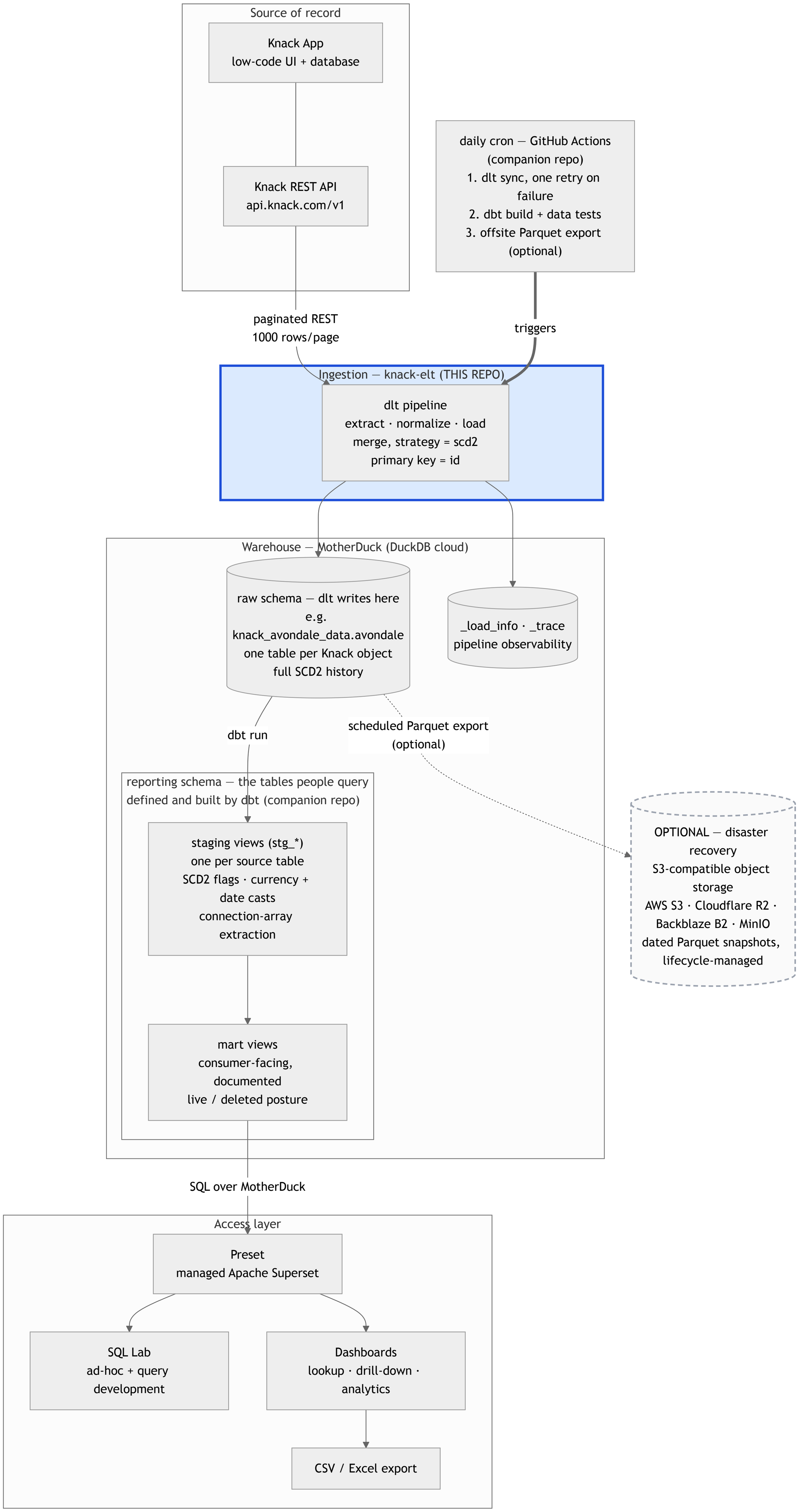


How to read it. The copy runs one way only — nothing this stack does can change data in Knack. The warehouse keeps history, so a record deleted in Knack is still answerable afterwards; that is the whole point of having it. The two boxes inside the warehouse are a division of labour, not two copies of the data: the complete history is what arrives, and **dbt** is the tool that turns it into the cleaned, business-shaped tables people actually query.

The dashed box is genuinely optional. The warehouse is already a second copy of Knack, so this is a third — insurance against losing the *warehouse* (a bad migration, an account problem, a provider outage), not against losing Knack. Add it when the warehouse becomes the only remaining copy of pruned records; skip it while Knack still holds everything. Any S3-compatible store works, and retention is normally handled by the bucket's own lifecycle rules rather than by code.

The same stack, with the technical detail

FIGURE 2 — THE SAME STACK, WITH THE TECHNICAL DETAIL



The load-bearing detail: schema ownership is split.

Schema	Owner	Contents	Rule
raw (e.g. `avondale`)	dlt	One table per Knack object, full SCD2 history, Knack-shaped values	**dbt never builds into it.** It is the pipeline's output, and a `dbt run` writing here would be clobbered on the next sync.
`reporting`	dbt	Staging + mart views	Everything downstream reads here. No dashboard queries raw directly.

Staging exists so that exactly one layer absorbs the mess Knack and dlt produce together — currency strings with commas, dates wrapped in JSON, connection fields as JSON arrays, and dlt's type-variant columns (when a value doesn't fit the type inferred from the first batch, dlt NULLs the base column and diverts the value to a sibling like `amount__v_double`, which makes a naive `SUM()` under-report silently). Marts and dashboards then get to be simple.

Why Preset: it is managed Superset, so SQL Lab doubles as the query-development environment and the dashboard builder against the same MotherDuck connection — a proven query becomes a dataset becomes a chart with no rewrite. Any BI tool with a DuckDB/MotherDuck SQLAlchemy connection substitutes here; the contract is the `reporting` schema, not the tool. Note that row-level security and embedded analytics are paid-tier features in Preset — untrusted multi-tenant self-service needs a different surface (a scoped API in front of MotherDuck), not a BI seat.

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2. Pipeline flow

What one run of knack-elt does. Two inputs from Knack: the **app metadata** (schema) and the **records** (data). The metadata is what turns `field_73` into `payroll_name`.


knack-elt run-pipeline
Schema pass — mapping.py
Knack app metadata
GET /applications/{app_id}
via knack-sleuth
create_app_mappings()
object_mappings
object name → object_id
numeric_fields
number · currency · link · date_time auto_increment · count
default_values
boolean field defaults
field_mappings
object_id → field_key → slug
Resource build — knack_dlt.py
for each Knack object
dlt.resource — get_knack_table_data()
client.paginate
/objects/{object_id}/records
rows_per_page=1000 · format=raw
stamp table_name + object_id
onto every row
row has an id?
no
yes
log warning, drop row
clean_json_fields()
invalid or empty JSON → None
chained with the pipe operator
dlt.transformer — get_remap_transformer()
clean_empty_strings()
empty string → None
in numeric fields
assign_default_values()
None or empty → declared default
remap_keys()
field_43 → event_name
Load — dlt
normalize
max_table_nesting=0 → flat tables
merge, strategy = scd2
primary_key = id
destination
DuckDB or MotherDuck
load_info + trace
→ _load_info / _trace tables

Notes on the mapping pass

- Field names are slugified with `re.sub(r'[^a-z0-9]+', '_', name.lower()).strip('_')`, so `"Vocalist / Song Details"` becomes `vocalist_song_details`.
- A user-defined Knack field literally named `id` would collide with Knack's own record id, so it is renamed to `<singular>_id` (e.g. `worker_id`). Knack's row id is always `id`, and it is the pipeline's primary key.
- `numeric_fields` and `default_values` are registered under the field key, the raw name, **and** the slug, because cleaning runs before and after the remap.
- `remap_keys` falls back to the original key when no mapping exists, so an unmapped object still loads — with `field_NN` column names.

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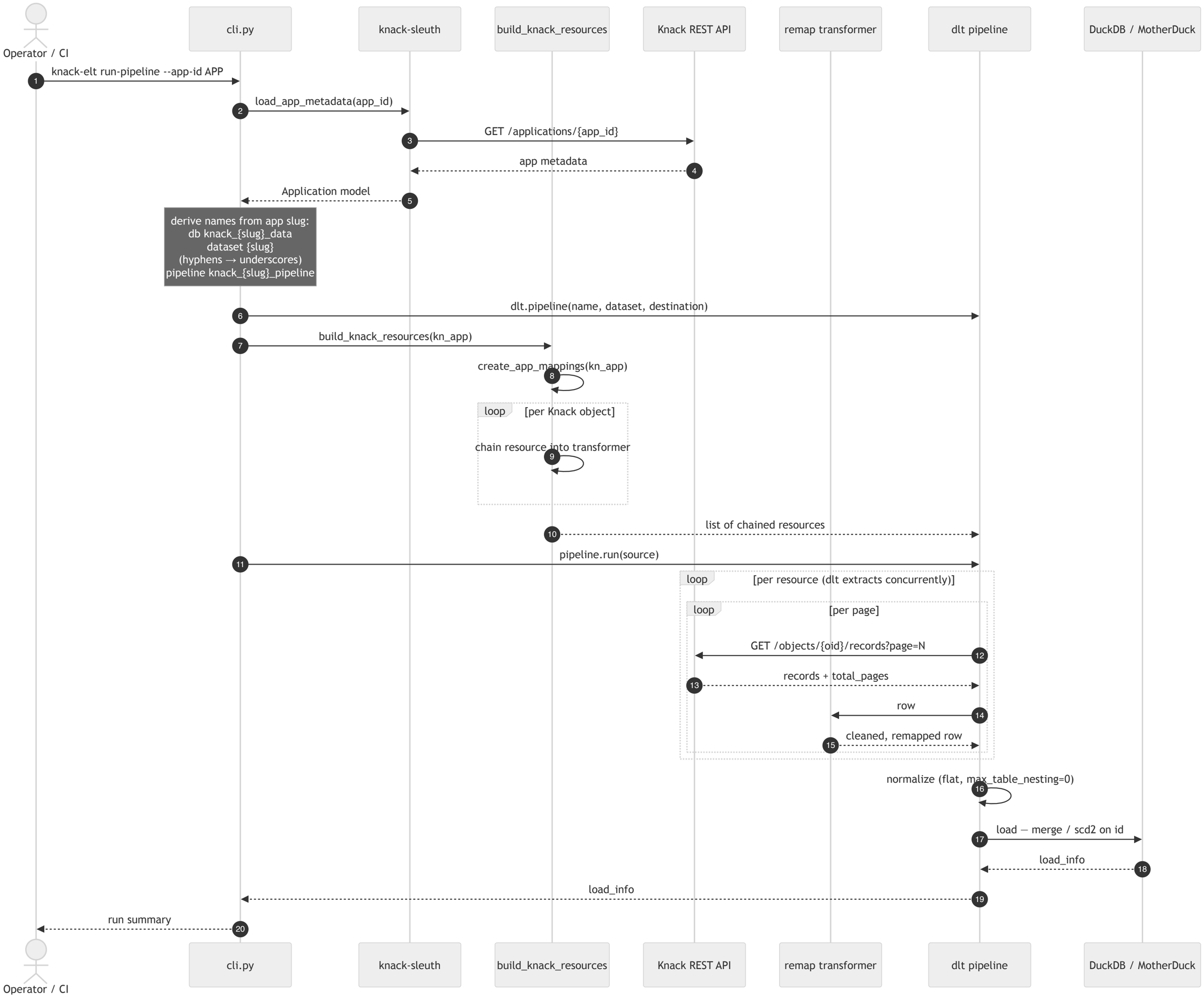
3. Run sequence

The packaged entry point is the Typer CLI (`knack-elt run-pipeline`). It takes the app metadata from knack-sleuth rather than fetching it itself.

FIGURE 4 — RUN SEQUENCE



Two entry points, and they do not target the same destination today:

Entry point	Metadata source	Object list	Destination
`cli.py run_pipeline` (`knack-elt run-pipeline`)	knack-sleuth `load_app_metadata()`	every object in the app	**local DuckDB** at `tests/data/knack_{slug}_data.duckdb` — the MotherDuck destination is present but commented out (`cli.py:80-84`, see the TODO at `cli.py:68`)
`python -m knack_elt.knack_dlt` (`knack_ave_source`)	its own HTTP `GET /applications/{id}`	every object in the app	**MotherDuck**, database name hardcoded to `knack_avondale_data`

The `__main__` path is the older one; it also sets `load.workers = 3`, `truncate_staging_dataset = True`, exports the schema to `pipelines/schemas/export`, and writes `load_info` and the run trace back to the destination as `_load_info` and `_trace`. Making the destination a CLI flag is the obvious next step and the TODO already names it.

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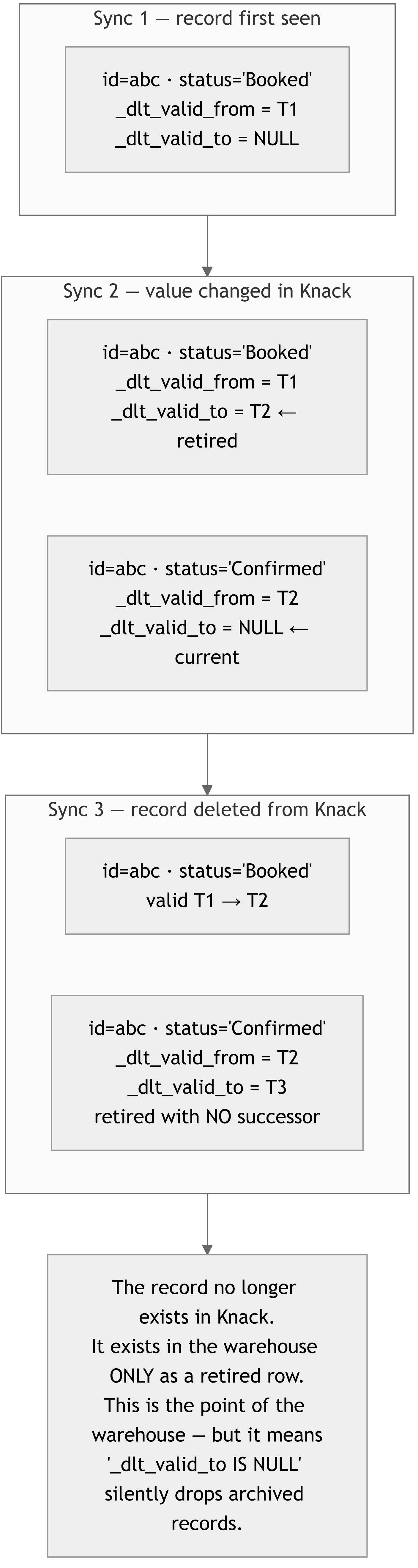
4. SCD2 row lifecycle

Every table is loaded with `write_disposition={"disposition": "merge", "strategy": "scd2"}`, so the warehouse is **append-and-retire, not overwrite**. dlt adds `_dlt_valid_from` and `_dlt_valid_to` to every row. A record's history is therefore several rows sharing one `id`.

FIGURE 5 — SCD2 ROW LIFECYCLE



Because of that last case, two different flags are needed, and conflating them is the most common way to get a wrong answer out of this stack:

```
-- one row per record, live or deleted, plus the live/deleted flag
with flagged as (
    select
        *,
        row_number() over (partition by id order by _dlt_valid_from desc) = 1
            as latest_version,
        _dlt_valid_to is null as is_live_in_knack
    from avondale.some_table          -- the raw, dlt-owned schema
)
select * from flagged where latest_version
```

Question	Filter
Current state of everything still in Knack	`latest_version and is_live_in_knack`
Latest known state of every record ever seen, including deleted	`latest_version`
What did this record look like on a given date	`_dlt_valid_from <= d and (_dlt_valid_to is null or _dlt_valid_to > d)`
What has been deleted from Knack	`latest_version and not is_live_in_knack`

Two traps worth stating explicitly:

- **Partition by `id`, not by a Knack field that looks like an id.** `id` is the canonical Knack record id supplied by the API and is populated on every row. A user-added "Record ID" *field* is NULL on rows created before it existed, and partitioning on it collapses all of them into one NULL group and corrupts the flags.
- **Aggregate without `latest_version` and you double-count.** Every historical version of a row is still a row. A `SUM()` over the raw table sums every version of every record.

Deleted-in-Knack is not the same as *archived by policy* — distinguishing an intentional retention prune from incidental deletion needs a separate manifest of what the pruning process removed; SCD2 alone cannot tell them apart.

Configuration

`pydantic-settings`, loaded from environment or `.env` (`src/knack_elt/config.py`):

Variable	Used for
`KNACK_APP_ID`	Knack application id — also the default for `--app-id`
`KNACK_API_KEY`	Knack REST API key (sent as `X-Knack-REST-API-Key`)
`motherduck_api_key`	MotherDuck token, interpolated into the `md:///` connection string

Note that dbt-duckdb reads the MotherDuck token from `motherduck_token`, not `motherduck_api_key` — the two layers disagree on the variable name, so a deployment running both needs to set or shim both.